

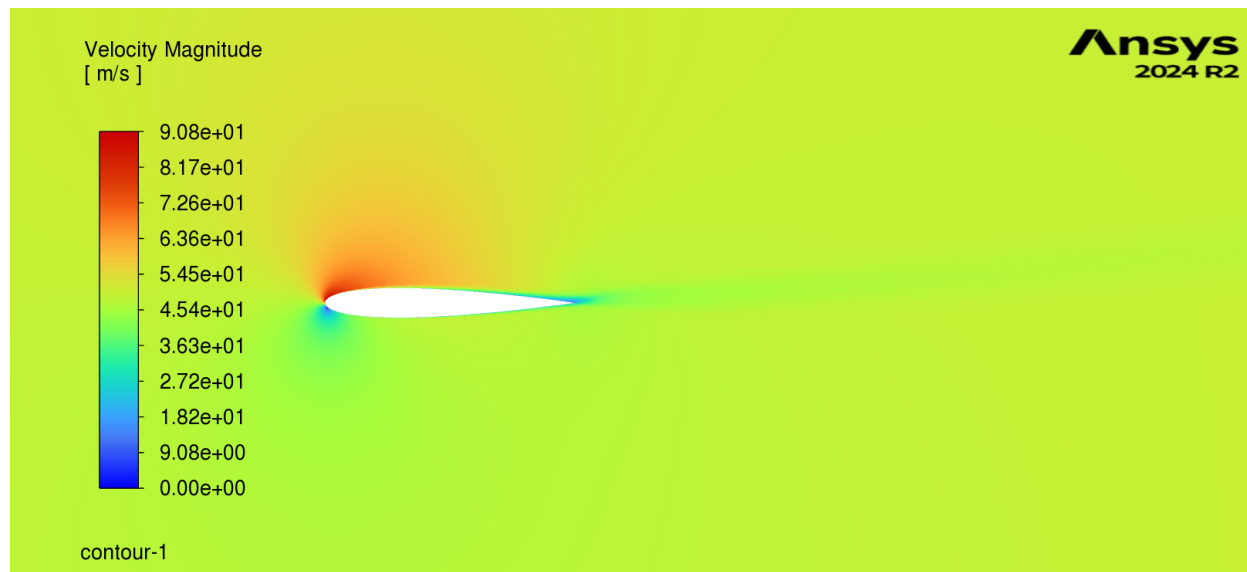
MENG-465 – Homework 7: Airfoil

November 6, 2025

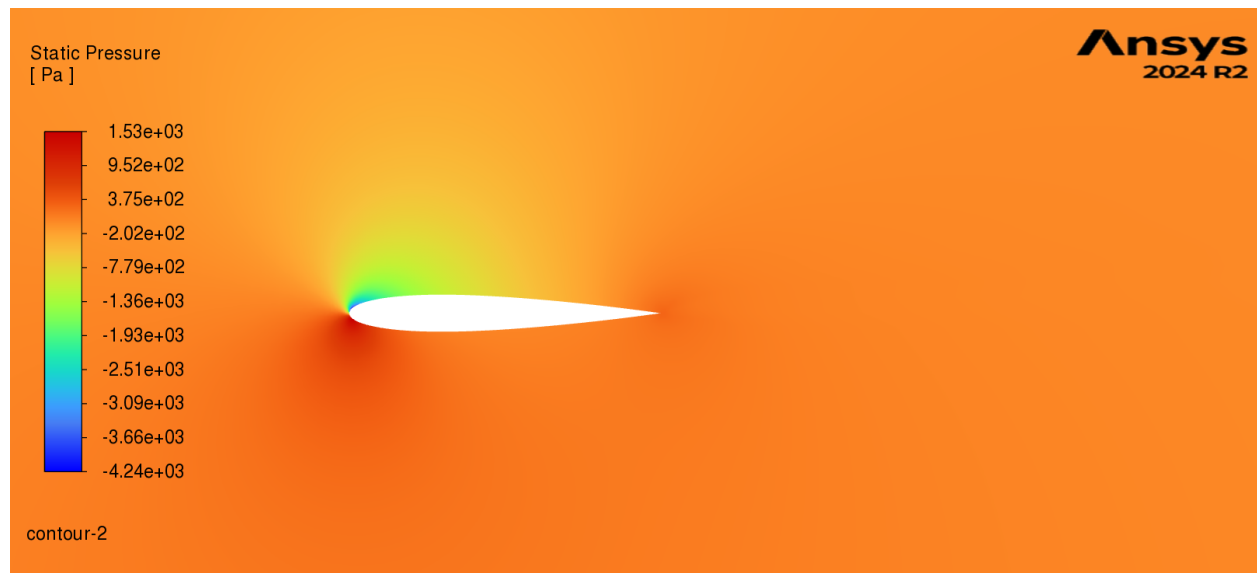
Derek Hissong

a. Angle of attack = 7deg

Velocity

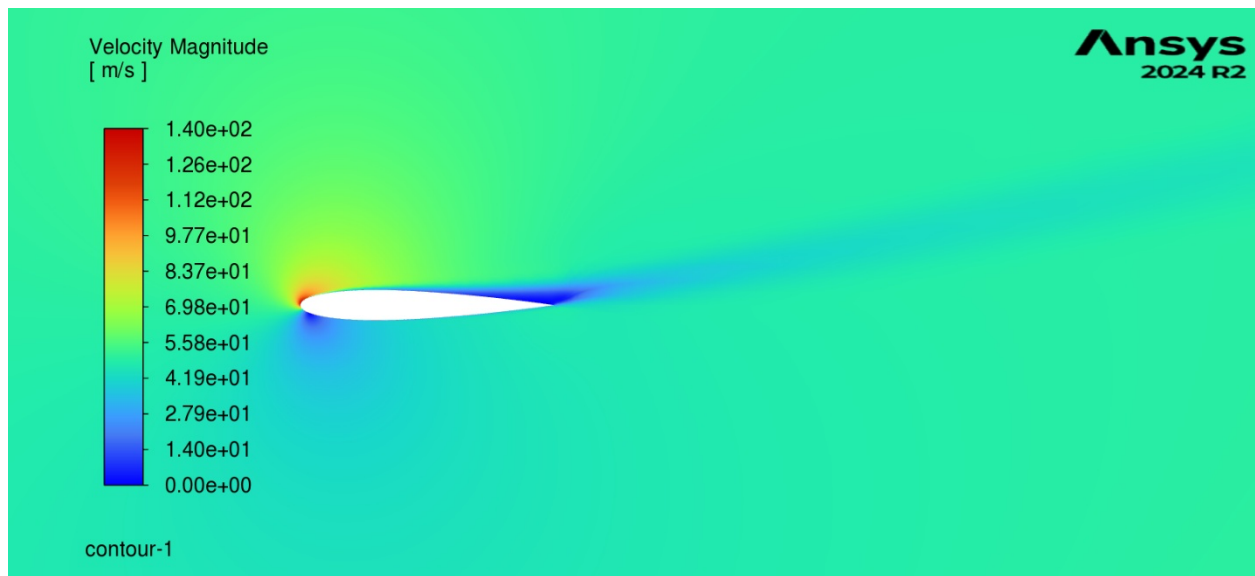


Static Pressure

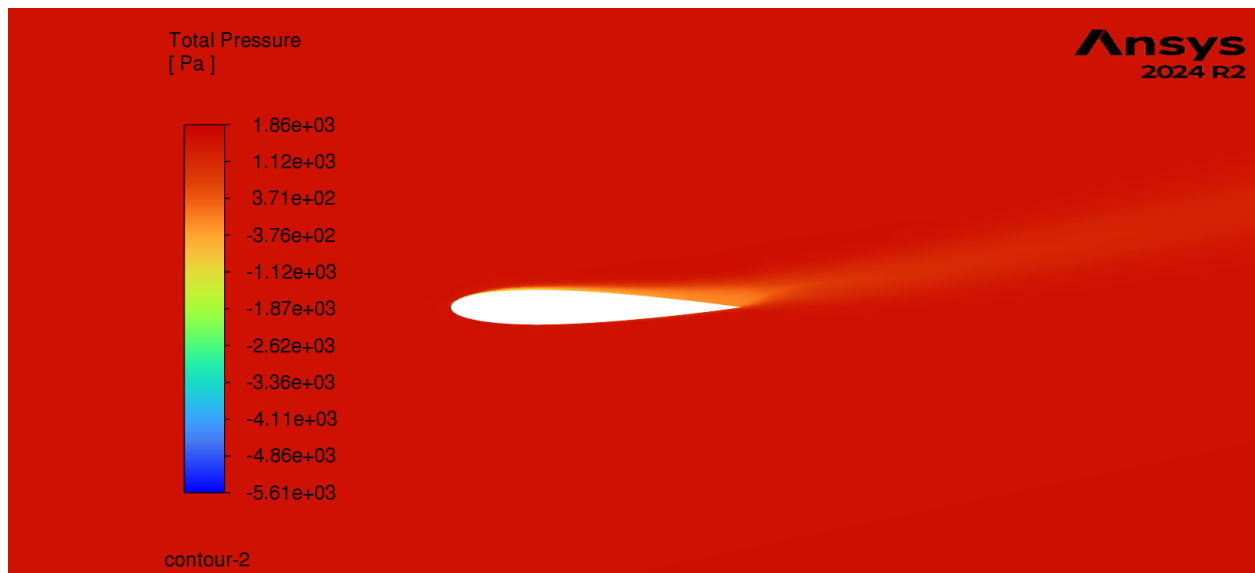


b. Angle of attack = 15deg

Velocity



Total Pressure



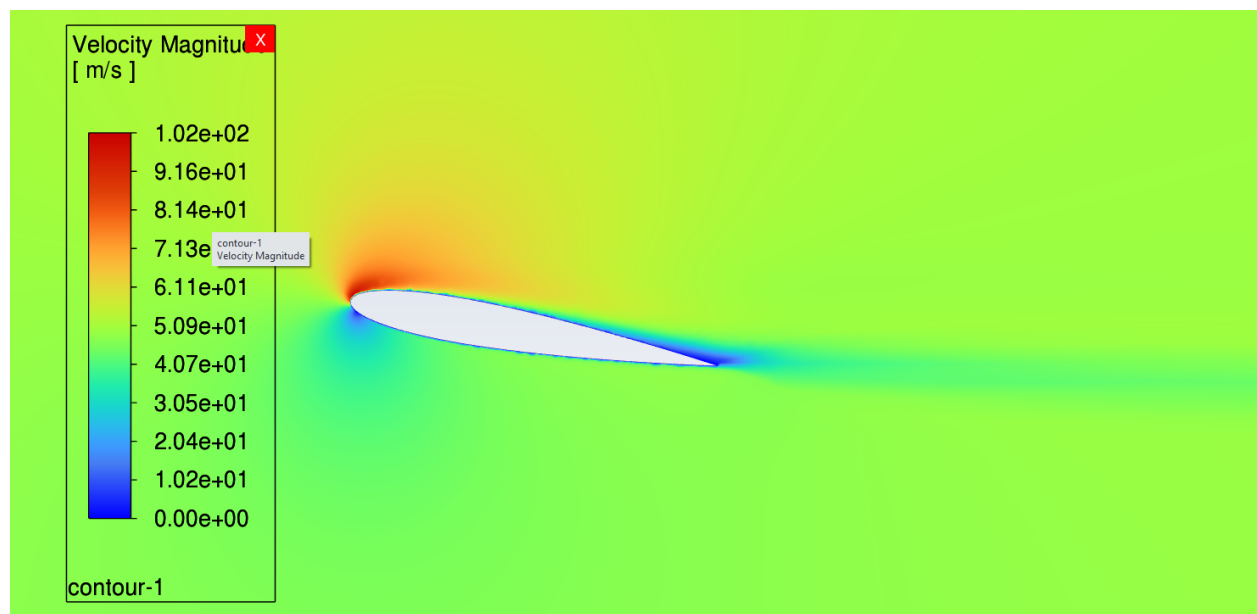
Verification:

$$D = C_D \left(\frac{1}{2} \right) \rho u^2 A ; A = \text{proj. cross area}$$

We kept getting negative drag force and coefficient, so we tried running the simulation tilting the airfoil off-axis and keeping the fluid traveling solely along the x-axis. When we ran the airfoil this way, we got positive numbers for the coefficient and drag force, but the coefficient is still wrong (so therefore so is the force) as it should not be so high. If we assume a projected area of 1m^2 or 1.5m^2 , then we should get a coefficient of 0.025 or 0.016, respectively.

This was done at a 10deg angle, Joe did one at 5deg.

Results are below:



Calculation complete.

Cd	
airfoil	62.990007
Drag [N]	
airfoil	38.58138

*Note, I worked on this with Joe in class!!